

Titanic Passenger Survival Analysis - Project Documentation

1. Project Overview

Analyze Titanic passenger survival data by manually inserting sample records into MySQL, to investigate survival patterns based on demographics, fare, and class.

2. Scope

- Design and create a MySQL database with a passengers table.
- Manually insert sample Titanic passenger data using SQL INSERT statements.
- Perform SQL queries to analyze survival statistics.
- No external dataset import involved.

3. Database Design & ER Diagram

- Table: passengers +-----+ | passengers | +-----+
+-----+ | PassengerID (PK) | | Survived | | Pclass | | Name | | Sex
| | Age | | SibSp | | Parch | | Ticket | | Fare | | Cabin | | Embarked |
+-----+ +

Column Data Type Constraints Description

PassengerID INT PRIMARY KEY Unique identifier for each passenger

Survived TINYINT NOT NULL Survival status (0 = No, 1 = Yes)

Pclass TINYINT NOT NULL Passenger class (1, 2, 3)

Name VARCHAR(255) NOT NULL Passenger name

Sex VARCHAR(10) NOT NULL Gender

Age FLOAT Age in years

SibSp TINYINT NOT NULL Number of siblings/spouses aboard

Parch TINYINT NOT NULL Number of parents/children aboard

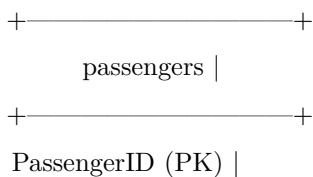
Ticket VARCHAR(50) NOT NULL Ticket number

Fare FLOAT NOT NULL Fare paid

Cabin VARCHAR(50) Cabin number

Embarked VARCHAR(10) Port of embarkation (C, Q, S)

ER Diagram (Text-based)



Survived |
Pclass |
Name |
Sex |
Age |
SibSp |
Parch |
Ticket |
Fare |
Cabin |
Embarked |
+-----+

4. Manual Data Insertion Sample

Sample SQL insert statements used to add passenger records:

```
INSERT INTO passengers (PassengerID, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked) VALUES
```

```
(1, 0, 3, 'Braund, Mr. Owen Harris', 'male', 22, 1, 0, 'A/5 21171', 7.25, NULL, 'S'),
```

```
(2, 1, 1, 'Cumings, Mrs. John Bradley (Florence Briggs Thayer)', 'female', 38, 1, 0, 'PC 17599', 71.2833, 'C85', 'C'),
```

```
(3, 1, 3, 'Heikkinen, Miss. Laina', 'female', 26, 0, 0, 'STON/O2. 3101282', 7.925, NULL, 'S');
```

5. Key SQL Queries for Analysis

View all inserted records:

```
SELECT * FROM passengers;
```

Count survivors:

```
SELECT Survived, COUNT(*) AS total FROM passengers GROUP BY Survived;
```

Survival rate by gender:

```
SELECT Sex, AVG(Survived) AS survival_rate FROM passengers GROUP BY Sex;
```

Average age of survivors:

```
SELECT AVG(Age) AS average_age FROM passengers WHERE Survived = 1;
```

Survival count by passenger class:

```
SELECT Pclass, COUNT(*) AS survived_count FROM passengers WHERE Survived = 1 GROUP BY Pclass;
```

6. Benefits and Learning Outcomes

- Understanding of manual data entry using SQL INSERT statements.
- Familiarity with database design and querying for data analysis.
- Practical demonstration of survival analysis with sample real-world data.

7. Future Enhancements

- Expand manual records for richer dataset.
- Introduce dataset import when accessible.
- Integrate visualization tools or front-end applications.